eLife

Reviewed Preprint
v2 • May 14, 2025
Revised by authors

Reviewed Preprint
v1 • February 26, 2025

Structural Biology and Molecular Biophysics

Structural mechanisms of PIP_2 activation and SEA0400 inhibition in human cardiac sodium-calcium exchanger NCX1

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eLife Assessment

Cardiac Ca2+/Na+ exchange is mediated by the NCX1 antiporter, whose activity is tightly regulated. This **important** manuscript describes the structural basis of activation by the lipid DiC8-PIP2 and inhibition by binding of a small molecule to NCX1. These results provide **convincing** insights into NCX1 regulation and the structural basis of cellular Ca2+ signaling.

https://doi.org/10.7554/eLife.105396.2.sa3

Abstract

Na^+/Ca^{2+} exchangers (NCXs) transport Ca^{2+} across the plasma membrane in exchange for Na^+ and play a vital role in maintaining cellular Ca^{2+} homeostasis. Our previous structural study of human cardiac NCX1 (HsNCX1) reveals the overall architecture of the eukaryotic exchanger and the formation of the inactivation assembly by the intracellular regulatory domain that underlies the cytosolic Na^+-dependent inactivation and Ca^{2+} activation of NCX1. Here we present the cryo-EM structures of HsNCX1 in complex with a physiological activator phosphatidylinositol 4,5-bisphosphate (PIP_2), or pharmacological inhibitor SEA0400 that enhances the inactivation of the exchanger. We demonstrate that PIP_2 binding stimulates NCX1 activity by inducing a conformational change at the interface between the TM and cytosolic domains that destabilizes the inactivation assembly. In contrast, SEA0400 binding in the TM domain of NCX1 stabilizes the exchanger in an inward-facing conformation that facilitates the formation of the inactivation assembly, thereby promoting the Na^+-dependent inactivation of NCX1. Thus, this study reveals the structural basis of PIP_2 activation and SEA0400 inhibition of NCX1 and provides some mechanistic understandings of cellular regulation and pharmacology of NCX family proteins.

Introduction

Sodium-calcium exchangers (NCX) are transporters that control the flux of Ca^{2+} across the plasma membrane and play a vital role in maintaining cellular calcium homeostasis for cell signaling[1–6]. NCXs facilitate the exchange of 3 Na^+ for 1 Ca^{2+} in an electrogenic manner, primarily responsible for extruding Ca^{2+} from the cytoplasm. However, this process can be reversed to permit Ca^{2+} entry, depending on the chemical gradients of Na^+ and Ca^{2+}, as well as the membrane potential[3,7–13]. Three NCX isoforms (NCX1-3) are present in mammals, with each isoform bearing multiple splice variants expressed in distinct tissues, thereby modulating numerous fundamental physiological events[4,14–19]. Dysfunctions of NCXs are associated with a plethora of human pathologies including cardiac hypertrophy, arrhythmia, and postischemic brain damage[3,20–22]. The cardiac variant NCX1.1 has been extensively studied, with its function playing a central role in cardiac excitation and contractile activity[23–27].

The eukaryotic NCX consists of a transmembrane (TM) domain with 10 TM helices and a large intracellular regulatory domain between TMs 5 and 6[4,28–33]. The TM domain is responsible for the ion exchange function in NCX. It consists of two homologous halves (TMs 1-5 and TMs 6-10) with TMs 2-3 and TMs 7-8 forming the core of the TM domain and hosting the residues that coordinate the transported Na^+ and Ca^{2+}[34,35]. The large regulatory domain contains two calcium-binding domains (CBD1 and CBD2). Ca^{2+} binding at CBDs enhances NCX activity and rescues it from the Na^+-dependent inactivation, a process that manifests as slow decay of the exchange current due to elevated levels of cytosolic Na^+ ions[12,30,31,36–41]. A stretch of residues known as XIP region (eXchanger Inhibitory Peptide) at the N terminus of the regulatory domain plays a pivotal role in the Na^+ inactivation process[42,43].

Several other cellular cues can also modulate NCX1 activity, including phosphatidylinositol 4,5-bisphosphate (PIP_2) in the membrane. PIP_2 has been shown to stimulate the NCX1 activity by reducing the Na^+-dependent inactivation and the XIP region was suggested to participate in the PIP_2 activation[39,44–46]. In addition, several small molecule NCX inhibitors have been developed to provide valuable tools for studying the physiological and pharmacological properties of NCXs, among which compound SEA0400 is a highly potent and selective NCX1 inhibitor that promotes the Na^+-dependent inactivation of the exchanger[20,47–52].

We previously determined the human cardiac NCX1 structure in an inward-facing inactivated state in which XIP and the β-hairpin between TMs 1 and 2 form a TM-associated four-stranded β-hub and mediate a tight packing between the TM and cytosolic domains, resulting in the formation of a stable inactivation assembly that blocks the TM movement required for ion exchange function[32]. The study also provides a mechanistic insight into how cytosolic Ca^{2+} binding at CBD2 destabilizes the inactivation assembly and activates the exchanger[32]. To expand our understanding of NCX modulation by PIP_2 lipid and small molecule inhibitors, we present the cryo-EM structures of human cardiac NCX1 in complex with PIP_2 or SEA0400, revealing the structural basis underlying PIP_2 activation and SEA0400 inhibition of NCX1.

Results

PIP_2 activation of NCX1

Phosphatidylinositol 4,5-bisphosphate (PIP_2) has been shown to modulate NCX1 activity by reducing the Na^+-dependent inactivation of the exchanger[39,44–46]. To characterize the effect of PIP_2 on HsNCX1, we expressed the exchanger in Xenopus laevis oocytes and recorded the outward exchanger currents using the giant patch technique in the inside-out configuration with or without applying porcine brain PIP_2 (**Fig. 1**, and Methods). The recording was performed with 12 μM free cytosolic $[Ca^{2+}]_i$ (bath) and the outward NCX1 current was elicited by the rapid replacement of 100 mM Cs^+ with 100 mM Na^+ in the bath solution. In the patches without applying PIP_2, the outward exchanger currents quickly decay and reach a steady state due to Na^+-dependent inactivation (**Fig. 1A**). Introducing 10 μM PIP_2 at the steady state progressively increases the current that plateaus at about two-fold of the peak current measured before PIP_2 addition (**Fig. 1A**). After PIP_2 washout, the outward current remains at the pre-removal level without obvious inactivation, indicating high-affinity PIP_2 binding and its positive modulation of NCX1 by both potentiating the peak current and reducing the Na^+-dependent inactivation. Intriguingly, the commonly used shorter chain PIP_2 substitute (PIP_2 diC8) does not have the equivalent activation effect on NCX1 and the exchanger remains susceptible to Na^+-dependent inactivation when recorded in the presence of 10 μM PIP_2 diC8 (**Fig. 1B**). However, PIP_2 diC8 still binds and stimulates both the peak and steady currents of the exchanger. This stimulation effect is abolished after PIP_2 diC8 washout, indicating a lower affinity of short-chain PIP_2 than that of the long-chain native PIP_2.

To verify that the short-chain PIP_2 diC8 and the long-chain brain PIP_2 share the same binding site, we performed a competition assay. As shown in **Fig. 1C**, introducing high-affinity brain PIP_2 at the steady state yields a long-lasting potentiation of NCX1 current that is irreversible even after a 5-minute washout. Applying PIP_2 diC8 can steadily decrease the brain PIP_2-potentiated NCX1 current, suggesting that both lipids compete for the same binding site.

Structural insight into PIP_2 binding in NCX1

To reveal the structural mechanism of PIP_2 activation, we tried to obtain the EM structure of HsNCX1 in the presence of the long-chain porcine brain PIP_2. However, the exchanger becomes highly dynamic, yielding a low-resolution EM map with an overall shape similar to a cytosolic Ca^{2+}-activated NCX1 whose β-hub-mediated inactivation assembly is destabilized and cytosolic domain (CBD1 and CBD2) is detached from the TM domain[32] (**Fig. S1**). We suspect the long-chain PIP_2 exerts the same activation effect on NCX1 as high cytosolic Ca^{2+} by destabilizing the inactivation assembly. As NCX1 retains its Na^+-dependent inactivation property in the presence of the short-chain PIP_2, we reasoned that the PIP_2 diC8-bound NCX1 likely remain in an inward-facing inactivated state in high Na^+ low Ca^{2+} condition and its structure would still reveal how PIP_2 binds in NCX1. We therefore determined the EM structure of NCX1 in complex with PIP_2 diC8 at 3.5Å (**Fig. 2**, **Fig. S2**, **Table S1**, and Methods), which indeed adopts an inward-facing conformation with intact inactivation assembly. Due to the relative dynamic movement between the TM and cytosolic domains, we also performed local refinement to improve the map quality for each domain (**Fig. S2**). The density of the IP_3 head group from the bound PIP_2 diC8 is well-defined in the local-refined EM map focused on the TM domain (**Fig. 2A** and **Fig. S2B**). This density is not present in the apo NCX1 structure (**Fig. S3**). The acyl chains, however, are flexible and could not be resolved in the structure (**Fig. S2**). The lipid is attached to the cytosolic sides of TMs 4 and 5 with its head group positioned at the C-terminal end of TM5 (**Fig. 2A**). Four positively charged residues including K164 and R167 from the N-terminus of TM4 and R220 and K225 from the C-terminus of TM5 are positioned in the vicinity of the PIP_2 head group and likely participate in the electrostatic interactions with the head group (**Fig. 2A**).

PIP_2 diC8-induced conformational changes in NCX1

Two major conformational changes occur in NCX1 upon PIP_2 diC8 binding (**Fig. 2B-D**). The first change occurs at the C-terminus of TM5 which ends at R220 and is connected to the 2-stranded XIP β-sheet (β3 and β4) via a 6-residue loop in the apo structure. When PIP_2 binds, the TM5 helix extends by one helical turn. This loop-to-helix transition significantly changes the locations of these connecting loop residues and their side-chain orientations to accommodate PIP_2, enabling

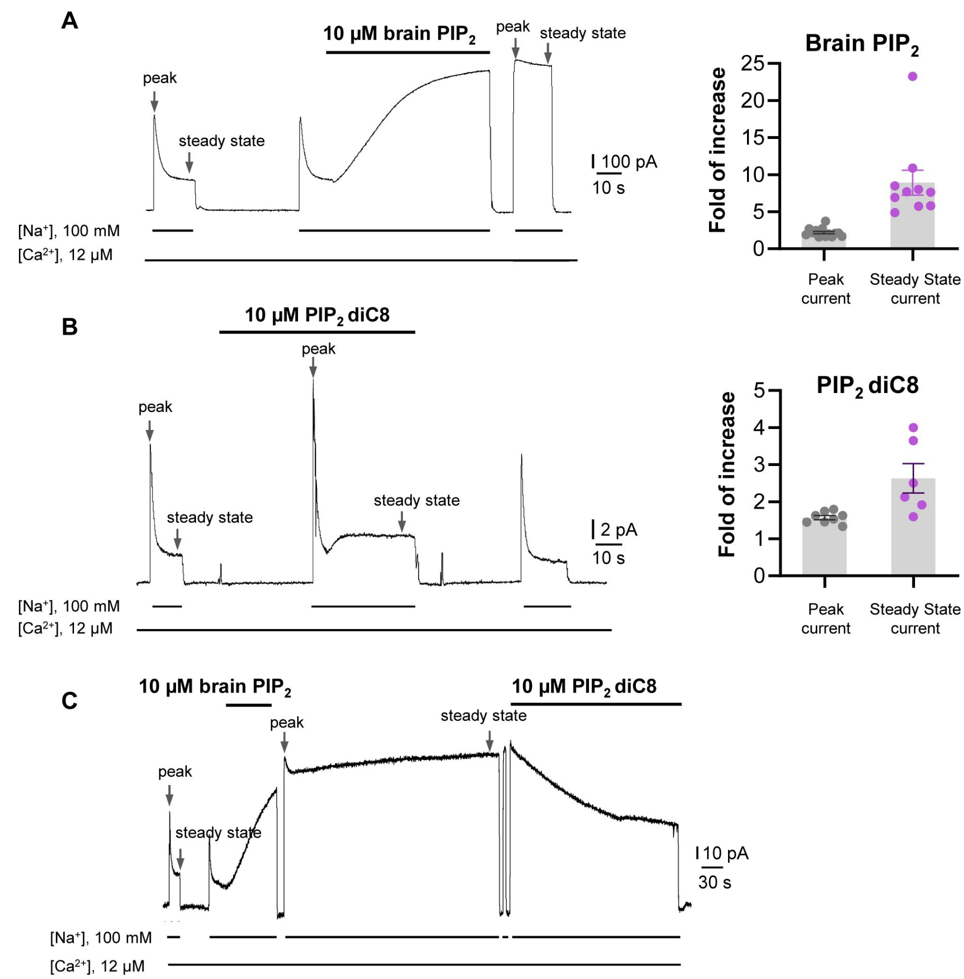

Fig. 1.

PIP_2 enhances HsNCX1 activity.

(A) Representative outward currents recorded from oocytes expressing the human NCX1 before and after application of long-chain brain PIP_2. Currents were activated by replacing cytosolic Cs+ with Na+. Application of 10 μM brain PIP_2 enhanced HsNCX1 current and abolished the Na+-dependent inactivation irreversibly. Perfusion time of PIP_2 is indicated above traces while lines below traces indicate solution exchange. Arrows mark the peak and steady currents used to measure the fold of increase upon PIP_2 application. The fold of current increase was calculated by comparing the peak or steady-state current before and after PIP_2 application.

(B) Representative outward currents recorded before and after application of short-chain PIP_2 diC8 (10 μM). PIP_2 diC8 was perfused from the cytosolic side before HsNCX1 activation (in the presence of Cs+ for 30 s) and during transport (in the presence of Na+). Both peak and steady-state currents of HsNCX1 are enhanced by PIP_2 diC8 and the effect is reversible. The Na+-dependent inactivation remains in the presence of PIP_2 diC8.

(C) Representative outward currents recorded with the application of brain PIP_2 and PIP_2 diC8. The NCX1 current was first potentiated by applying 10 μM brain PIP_2 at the steady state. The PIP_2 effect was not reversible over the 5-minute washout with a solution containing 100 mM Na+ and 12 μM Ca2+. The same patch was then perfused with the same solution in the presence of 10 μM PIP_2 diC8. Application of the short-chain PIP_2 diC8 facilitates the decrease of brain PIP_2-potentiated current, suggesting that both lipids compete for the same binding site.

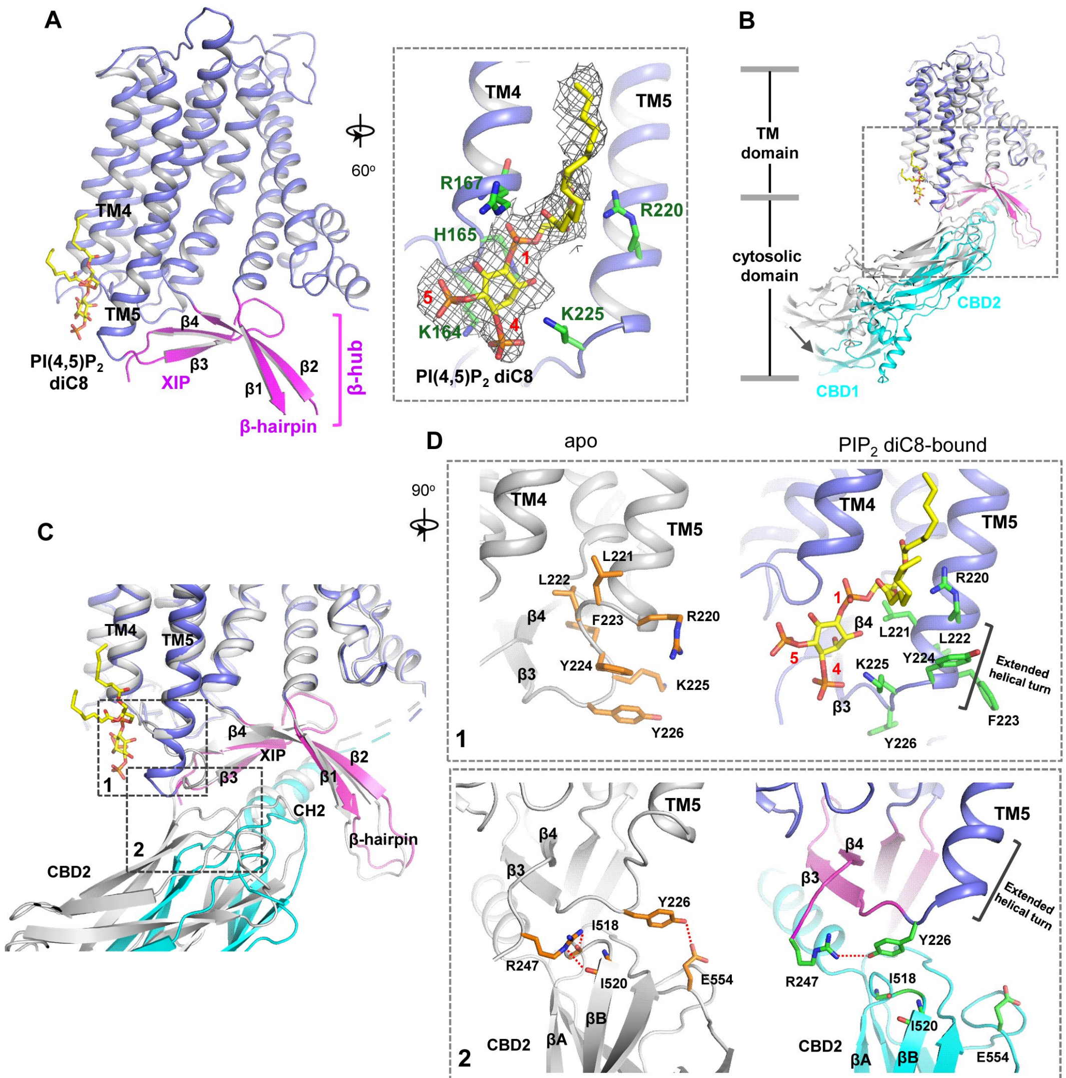

Fig. 2.

PIP$_2$ binding in NCX1.

(A) Structure of the TM and β-hub regions of PIP$_2$ diC8-bound NCX1 with a zoomed-in view of the lipid binding site. The density of PIP$_2$ diC8 is shown as a grey mesh contoured at 5.5σ.

(B) Structural comparison between apo (grey) and PIP$_2$ diC8-bound (color) NCX1. Upon PIP2 diC8 binding, there is a rigid-body downward swing movement (marked by an arrow) at CBDs caused by the partial detachment of the CBD2 domain from XIP. The conformational change at the TM domain is subtle and mainly occurs at the C-terminus of TM5 as illustrated in (**C**) and (**D**).

(C) Zoomed-in view of the structural comparison (boxed area in (B)). The two major conformational changes occur in the boxed regions. **(D)** Zoomed-in views of the two conformational changes between apo (left in grey) and PIP$_2$ diC8-bound (right in color) state. Top: conformational change 1 at the C-terminus of TM5. Bottom: conformational change 2 at the interface between XIP and CBD2.

K225 to reorient and interact with the IP_3 head group (**Fig. 2D**, top panel). The second conformational change is the partial detachment of the CBD2 domain from XIP upon PIP_2 binding, resulting in a downward swing of cytosolic CBD domains (CBD1 and CBD2) as a rigid body (**Fig. 2B&C**). This CBD2 detachment is a result of the first PIP_2-induced conformational change at TM5 that disrupts part of the interactions between CBD2 and XIP as follows: In the apo inactivated state, Y226 and R247, the two termini residues of the 2-stranded XIP β-sheet, form H-bonds with several CBD2 residues including E554 sidechain and backbone carbonyls of I518 and I520 (**Fig. 2D**, bottom panel); The loop-to-helix transition of TM5 upon PIP_2 binding leads to a dramatic rotation of Y226 that allows it to move closer to and directly interact with R247, resulting in the loss of H-bonding interactions between XIP and CBD2 (**Fig. 2D**, bottom panel); In addition, the rotation of Y226 also causes a direct collision with CBD2 if it remains closely attached to XIP. Thus, the PIP_2-induced TM5 movement, particularly the reorientation of Y226, abolishes some local interactions between CBD2 and XIP and also pushes CBD2 away from XIP, causing the partial detachment of CBD2. However, the short-chain PIP_2 only partially destabilizes rather than completely disassembles the inactivation assembly as the CH2 helix of CBD2 still engages in extensive interactions with the C-shaped β-hub.

To test if the PIP_2-interacting residues play a critical role in the native long-chain PIP_2 activation, we performed mutagenesis to those positively charged residues, including K164A, R167A, R220A, and K225A single mutants. We also generated an R220A/K225A double mutant as these two residues undergo PIP_2-induced conformational change at TM5. While all mutants remain susceptible to current potentiation upon brain PIP_2 application as seen in the wild-type NCX1, the PIP_2-potentiation effects on peak and steady-state currents are weakened in some mutants, most notably in R220A and R220A/K225A (**Fig. 3A-C**). Interestingly, these two mutants also have reduced Na^+-dependent inactivation as demonstrated by their higher fractional activity (ratio between stead-state and peak currents) before applying PIP_2 (**Fig. 3D**). The retained PIP_2 activation on these mutants implies that the longer acyl chain from native PIP_2 plays an important role in its interaction and activation of NCX1. Furthermore, as the interactions between PIP_2 and NCX1 are both electrostatic involving multiple charged residues and hydrophobic involving the long lipid acyl chain, those single or double amino acid substitutions may only decrease the affinity of PIP_2 rather than abolish its binding. To test that, we also mutated all four positively charged residues to alanine. The K164A/R167A/R220A/K225A mutant is no longer sensitive to PIP_2 activation. The currents from this quadruple mutant are small in most recordings and show no Na^+-dependent inactivation. The unresponsiveness to PIP_2 and lack of Na^+-dependent inactivation in this mutant is consistent with previous studies demonstrating that PIP_2 activates NCX by tuning the amount of Na^+-dependent inactivation and any mutation that decreases NCX sensitivity to PIP_2 will also affect the extent of Na^+-dependent inactivation[44]. This quadruple NCX1 mutant likely abolishes the potentiation effect from both endogenous and externally applied PIP_2 and thereby functions in a Na^+-inactivated steady state.

Structure of NCX1 in complex with SEA0400 inhibitor

SEA0400 is known to potently inhibit cardiac NCX1 by facilitating the inactivation of the exchanger[50–52]. To reveal the structural mechanism of its inhibition, we determined the structure of HsNCX1 in complex with SEA0400 at 2.9 Å (**Fig. 4**, **Fig. S4**, **Table S1**, and Methods). The SEA0400-bound NCX1 structure adopts an inward-facing, inactivated state identical to the apo NCX1 structure obtained at high Na^+, nominal Ca^{2+}-free condition [32] (**Fig. 4A** and **Fig. S5**). SEA0400 binds at the TM domain in a pocket enclosed by the internal halves of TMs 8 (8a segment), 2 (2ab segments), 4, and 5 (**Fig. 4B**). The pocket has a lateral fenestration in the middle of the membrane (**Fig. 4B**), which provides a portal for the SEA0400 entrance. The pocket is sealed off at the cytosolic side by E244 from the XIP β-sheet. As demonstrated in our previous study[32], XIP and the linker β-hairpin (β1 and β2) between TMs 1 and 2ab form a β-hub that stabilizes the exchanger in an inactivated state. This β-hub has to be dissembled in an activated NCX1 and XIP is expected to be detached from the TM domain, which would lead to the

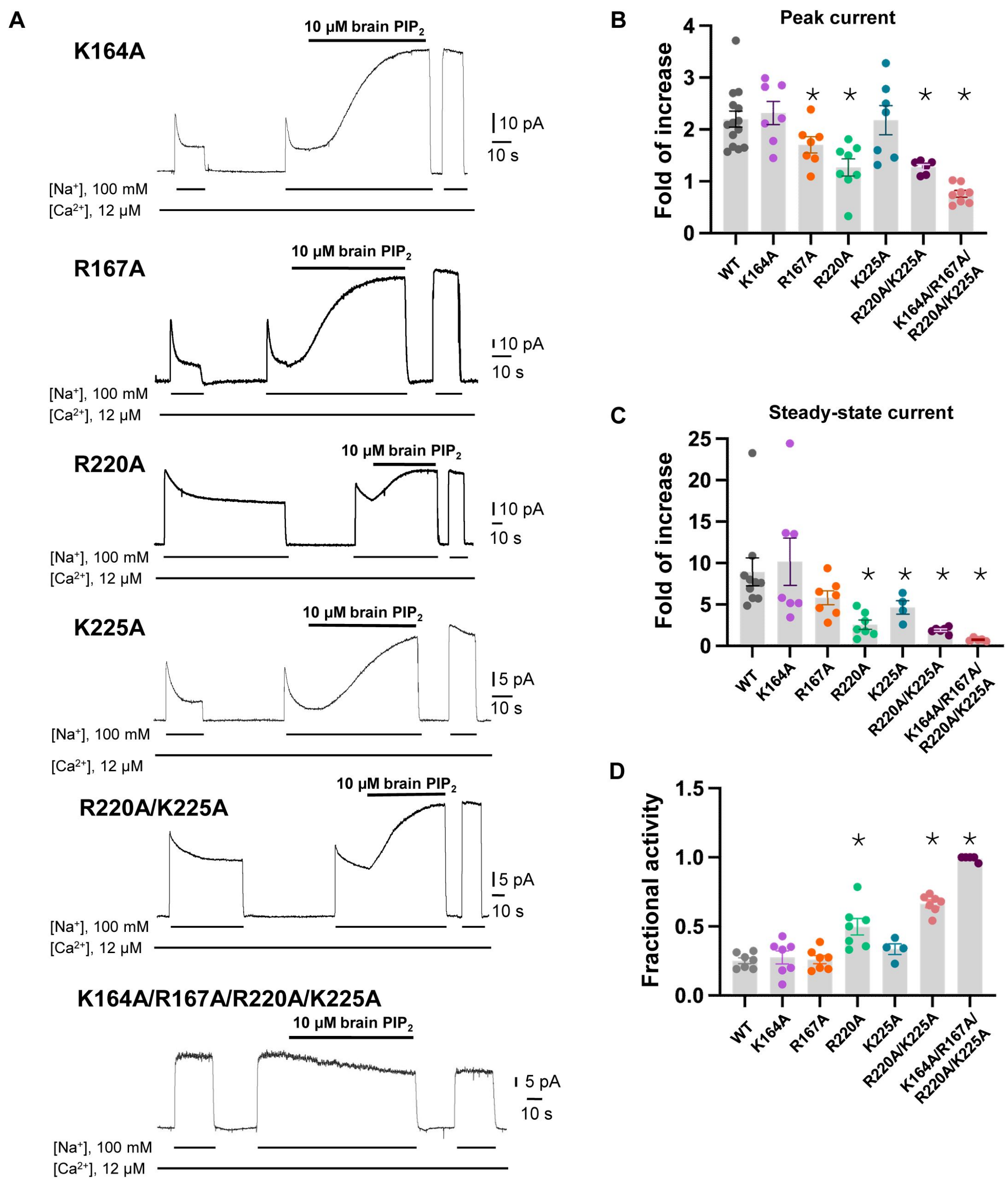

Fig. 3.

Mutagenesis at the PIP_2 binding.

(A) Representative NCX1 currents of PIP_2 site mutants before and after perfusion of 10 μM brain PIP_2 to the cytosolic side of the patch.

(B&C) Summary graphs demonstrating the effects of PIP_2 on the enhancement of peak (B) and steady-state currents (C). Potentiation (fold of increase) was measured by comparing the current magnitude before and after PIP_2 application. Mutants R167A, R220A, and K225A showed some decreased response to PIP_2 whereas R220A/K225A mutant shows more profound decrease in PIP_2 response. Compared to WT, the PIP_2 potentiation of R220A/K225A at the steady state is decreased by ~70-90% (Fold of increase WT=8.9±1.6, n=10 vs R220A/K225A=1.9±0.1, n=6). PIP_2 has no potentiation effect on the quadruple K164A/R167A/R220A/K225A mutant. Data points are mean ± s.e.m. (* $p < 0.1$).

(D) The extent of Na+-dependent inactivation was measured as the ratio between steady state and peak currents (fractional activity) and values for WT and the indicated mutants are shown. Mutants R220A and R220A/K225A displayed significantly higher fractional activity values when compared to WT, indicating that the Na+-dependent inactivation was less pronounced in these mutant exchangers. K164A/R167A/R220A/K225A shown no Na+-dependent inactivation.

opening of the SEA400-binding pocket to the cytosol and provide a cytosolic portal for the release of the inhibitor release as further discussed below. **Fig. 4C** summarizes the interactions between SEA0400 and NCX1 and demonstrates that SEA0400 fits perfectly in the pocket, making extensive contact with surrounding residues. Mutations of some key interacting residues, such as F213 and G833, have been shown to compromise the inhibitor binding[52]. The same SEA0400 binding was also demonstrated in the recent study of human NCX1.3 and mutations at some pocket-forming residues mitigate the SEA400 inhibition of the exchanger[33].

Inhibition mechanism of SEA0400

The TM domain of NCX1 shares a similar overall architecture to the archaea exchanger NCX_Mj[34]. The structural comparison between the TM domains of the inward-facing NCX1 and the outward-facing NCX_Mj reveals the conformational changes that occur during ion exchange, providing structural insight into the SEA0400 inhibition mechanism (**Fig. 5A**). The inward-outward transition mainly involves the sliding motion of TMs 1 and 6 and the bending movement of TMs 2ab and 7ab[32,34,35,53]. As TM2ab directly interacts with SEA0400 in the inward-facing state, its bending movement towards the outward conformation would cause a direct collision with the inhibitor (**Fig. 5A**). Thus, the binding of SEA0400 stabilizes the exchanger in the inward-facing state and blocks the conformational change from the inward to the outward state. The Na^+-dependent NCX1 inactivation occurs when the exchanger is in a Na^+-loaded, inward-facing state with low cytosolic $[Ca^{2+}]$[40,41,54]. As demonstrated in our previous study, only in this state, the β-hub can form and readily interact with the cytosolic CBD domains, generating the inactivation assembly that locks TMs 1 and 6 and prevents the TM module from transporting ions[32]. Thus, SEA0400 promotes NCX1 inactivation by stabilizing NCX1 in the inward-facing conformation which facilitates the formation of the inactivation assembly. The formation of the inactivation assembly also reciprocally stabilizes SEA0400 binding as the XIP of the assembly interacts with the TM domain and seals off the inhibitor binding pocket from the cytosolic side (**Fig. 5B**). Under conditions in which the inactivation assembly cannot form in NCX1, such as chymotrypsin treatment or forward exchange mode (Na^+ influx/Ca^{2+} efflux), the removal or detachment of XIP from the TM domain would generate a cytosolic open portal for SEA0400 release and effectively reduce its binding affinity (**Fig. 5C**)[50,51]. Indeed, cysteine scanning mutagenesis studies have shown that the pocket-forming G833 residue is accessible to intracellular sulfhydryl reagents in the chymotrypsin-treated inward-facing NCX1, confirming the cytosolic exposure of the SEA400-binding pocket upon XIP removal[55]. This cytosolic opening of the SEA0400 pocket explains the low efficacy of SEA0400 inhibition in NCX1 when the Na^+-dependent inactivation is absent.

Discussion

In this study, we provide mechanistic underpinnings of cellular regulation and pharmacology of NCX family proteins by revealing the structural basis of PIP_2 activation and SEA0400 inhibition in HsNCX1. Both compounds modulate NCX1 activity by reducing or enhancing the physiologically relevant Na^+-dependent inactivation process that occurs when the exchanger is in an inward-facing, Na^+-loaded state with high $[Na^+]$ and low $[Ca^{2+}]$ on the cytosolic side[27,40]. In this inactivation state, XIP and β-hairpin can assemble into a TM-associated four-stranded β-hub that mediates a tight packing between the TM and cytosolic domains, resulting in the formation of an inactivation assembly that blocks the TM conformational changes required for ion exchange function[32,33]. PIP_2 or SEA0400 binding changes the stability of the inactivation assembly in NCX1, resulting in a reduction or potentiation of inactivation.

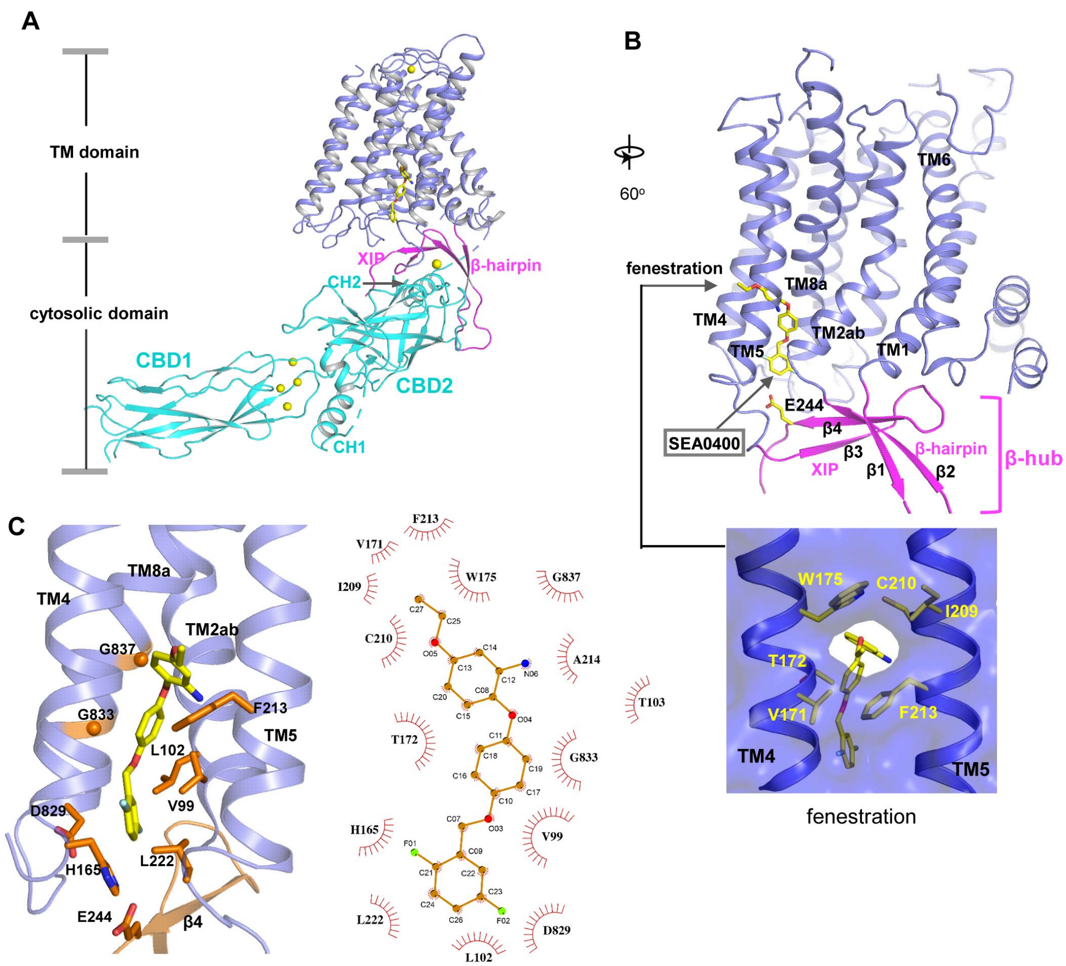

Fig. 4.

SEA0400 binding in NCX1.

(A) Overall structure of human NCX1 in complex with SEA0400 obtained in high Na+ and low Ca2+ conditions. Yellow spheres represent the bound Ca2+ in CBD1 and XIP.

(B) Cartoon representation of the TM domain and β-hub of the complex with surface-rendered view of the fenestration in the middle of the membrane. The β-hub is assembled by β-hairpin (β1 and β2) and XIP (β3 and β4).

(C) Zoomed-in view of SEA0400 binding site and the schematic diagram detailing the interactions between NCX1 residues and SEA0400.

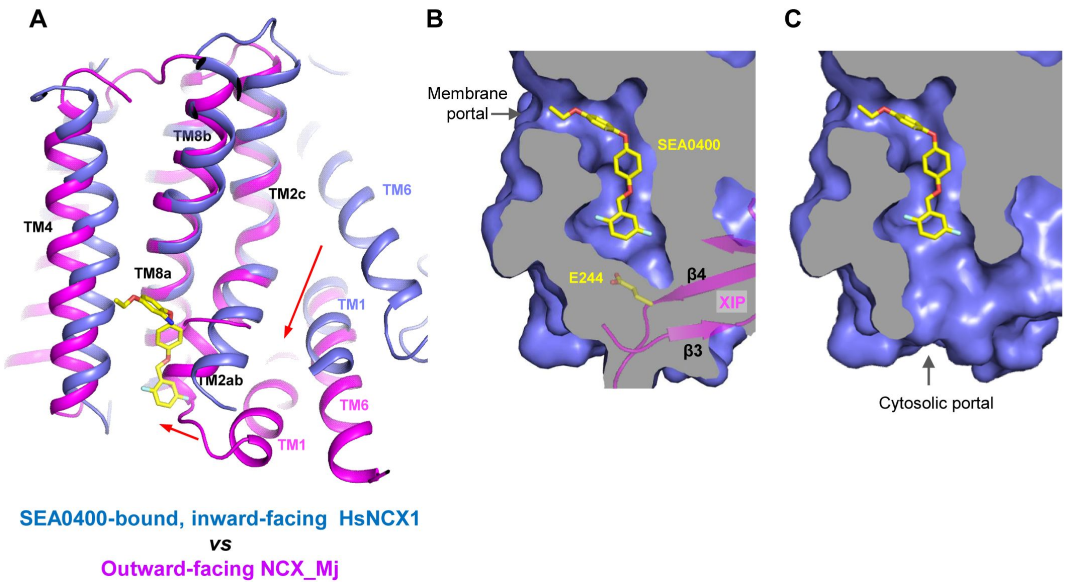

Fig. 5.

Structural mechanism of SEA0400 inhibition.

(A) Structural comparison at the core part of the TM domain between the SEA0400-bound, inward-facing HsNCX1 and the outward-facing NCX_Mj (PDB 3V5U). Red arrows mark the sliding movement of TMs 1 and 6 and the bending of TM2ab from inward to outward conformation.
(B) Surface-rendered views of the SEA0400-binding pocket sealed off from the cytosolic side by E244 from XIP in the inactivated state.
(C) Removal of XIP would generate a cytosolic portal that facilitates the release of SEA0400.

Although short-chain PIP_2 cannot fully recapitulate the NCX1 activation effect from long-chain native PIP_2, the structure of PIP_2 diC8 NCX1 allows us to define the lipid-binding site and the lipid-induced conformational changes at the interface between CBD2 and XIP that destabilize the inactivation assembly. The SEA0400-bound NCX1 structure presented here, along with the recent study by Dong et al.[33], suggests that the drug can directly inhibit NCX1 by blocking the inward-outward conformational change at TM2ab. Our structural analysis also explains the strong connection between SEA0400 binding and the Na^+-dependent inactivation - SEA0400 is ineffective in an exchanger lacking Na^+-dependent inactivation whereas enhancing the extent of Na^+-dependent inactivation increases the affinity for SEA0400. As SEA0400 binding traps NCX1 in an inward-facing state that facilitates the formation of the inactivation assembly, the interaction between XIP and the TM domain in the assembly can in turn stabilize SEA0400 binding by sealing the drug-binding pocket from the cytosol. When XIP is removed as in the chymotrypsin-treated NCX1 or when the inactivation assembly cannot form as in the NCX1 at forward exchange mode, the SEA0400-binding pocket becomes exposed to the cytosol, mitigating the inhibition efficacy of SEA0400[50,51].

While we expect the long-chain PIP_2 to bind in the same location as PIP_2 diC8, it is unclear how it exacerbates the destabilization effect on the inactivation assembly. The long acyl chain of native PIP_2 may engage in some interactions with the TM module of NCX1 that are not present in the short-chain lipid, rendering a higher affinity binding and more profound destabilization effect on the inactivation assembly than PIP_2 diC8. The acyl-chain length-dependent PIP_2 activation is consistent with some previous studies. Before PIP_2 was demonstrated to regulate NCX, some earlier studies showed that negatively charged long-chain lipids such as phosphatidylserine (PS) or phosphatidic acid (PA) could have the same potentiation effects on NCX1 as PIP_2[56,57]. Furthermore, long-chain acyl-CoAs could also have the same potentiation effects on NCX as PIP_2[58]. All these studies demonstrated that activation of NCX by the anionic lipids depends on their chain length with the short chain being ineffective or less effective. These findings have two implications. First, it is the negative surface charge rather than the specific IP_3 head group of the lipid that is important for stimulating NCX1 activity, suggesting non-specific electrostatic interactions between the negatively charged lipids and those positively charged residues at the binding site. Second, a longer acyl chain is required for the high-affinity binding of PIP_2 or negatively charged lipids. In the PIP_2 diC8-bound structure, the tail of the acyl chain is positioned right at the fenestration of NCX1 that serves as the portal for the SEA0400 binding. Two pieces of evidence lead us to suggest that the tail of the long acyl chain from a native lipid can enter the same binding pocket for SEA0400 and thereby rendering its higher affinity binding than a shorter chain lipid. Firstly, a docking analysis[59,60] of both long-chain and short-chain PIP_2 at the binding site showed that the tail portion of the acyl chain from the native brain PIP_2 inserts into the SEA0400-binding pocket through the fenestration of NCX1 in all docking models with highest calculated affinity (**Fig. S6A**). This pocket is not reachable for PIP_2 diC8 due to its shorter chain length (**Fig. S6B**). Second, a stretch of density likely from a native lipid acyl chain is observed in the apo NCX1 structure inside the SEA0400-binding pocket (**Fig. S6C**), indicating its accessibility to a long-chain lipid. Thus, the accommodation of the long acyl chain in the open pocket of NCX1 through the fenestration in the middle of the membrane likely contributes to the high affinity binding of native lipids.

Methods

Protein expression and purification

The expression and purification of the HsNCX1 (cardiac isoform NCX1.1, indicated as HsNCX1 or NCX1 throughout the manuscript) were carried out as described previously[32]. Truncated HsNCX1 (Δ341-365aa) containing a C-terminal Strep-tag was cloned into a pEZT-BM vector and baculoviruses were produced in *Sf9* cells[61]. For protein expression, cultured Expi293F GnTI-cells were infected with the baculoviruses at a ratio of 1:20 (virus: GnTI-, v/v) for 10 hours. 10 mM sodium butyrate was then introduced to boost protein expression level and cells were cultured in suspension at 30°C for another 60 hours and harvested by centrifugation at 4,000 g for 15 mins. All purification procedures were carried out at 4°C. The cell pellet was resuspended in lysis buffer (25 mM HEPES pH 7.4, 300 mM NaCl, 2 μg/ml DNase I, 0.5 μg/ml pepstatin, 2 μg/ml leupeptin, 1 μg/ml aprotinin, and 0.1 mM PMSF) and homogenized by sonication. NCX1 was extracted with 2% (w/v) N-dodecyl-β-D-maltopyranoside (DDM, Anatrace) supplemented with 0.2% (w/v) cholesteryl hemisuccinate (CHS, Sigma Aldrich) by gentle agitation for 2 hours and supernatant collected by centrifugation at 40,000 g for 30 mins was incubated with Strep-Tactin affinity resin (IBA) for 1 hour. The resin was then collected on a disposable gravity column (Bio-Rad) and washed with 30 column volumes of buffer A (25 mM HEPES pH 7.4, 200 mM NaCl, 0.06% digitonin). NCX1 was eluted in buffer A supplemented with 50 mM biotin and further purified by size-exclusion chromatography on a Superdex200 10/300 GL column (GE Healthcare). For the generation of NCX1-Fab 2E4 complex, NCX1 was incubated with purified Fab in a molar ratio of 1:1.2 (NCX1: Fab 2E4) for 2 hours and further purified by size-exclusion chromatography in buffer A. The peak fractions were collected and concentrated to ~5-6 mg/ml for cryo-EM analysis. To prepare the protein samples in complex with various compounds, 0.9 mM SEA0400, 0.47 mM PI(4,5)P_2 diC8 or 0.42 mM brain PI(4,5)P_2 were added to the protein samples 2 hours before grid preparation.

Expi293F GnTI-cells were purchased from and authenticated by Thermo Fisher Scientific. The cell lines were tested negative for mycoplasma contamination.

Cryo-EM sample preparation and data acquisition

HsNCX1-Fab 2E4 samples (~5-6 mg/ml) in various conditions were applied to a glow-discharged Quantifoil R1.2/1.3 300-mesh gold holey carbon grid (Quantifoil, Micro Tools GmbH, Germany), blotted for 4.0 s under 100% humidity at 4 °C and plunged into liquid ethane using a Mark IV Vitrobot (FEI). For the SEA0400-bound NCX1-Fab 2E4, raw movies were acquired on a Titan Krios microscope (FEI) operated at 300 kV with a K3 camera (Gatan) at 0.83 Å per pixel and a nominal defocus range of −0.9 to −2.2 μm. Each movie was recorded for about 5 s in 60 subframes with a total dose of 60 $e^-/Å^2$. For other samples, raw movies were acquired on a Titan Krios microscope operated at 300 kV with a Falcon 4i (Thermo Fisher Scientific) at 0.738 Å per pixel and a nominal defocus range of −0.8 to −1.8 μm. Each movie was recorded for 4 s with a total dose of 60 $e^-/Å^2$.

Image processing

Cryo-EM data were processed following the general scheme described below with some modifications to different datasets (Figs. S1, S2, and S4). First, movie frames were motion-corrected and dose-weighted using MotionCor2[62]. The CTF parameters of the micrographs were estimated using the GCTF program[63]. After CTF estimation, micrographs were manually inspected to remove images with bad defocus values and ice contamination. Particles were picked using program Gautomatch (Kai Zhang, *https://www.mrc-lmb.cam.ac.uk/kzhang/*) or crYOLO[64] and extracted with a binning factor of 3 in RELION[65,66]. Extracted particles were subjected to 2D classification, ab initio modeling, and 3D classification. The particles from the best-resolving 3D class were then re-extracted with the original pixel size and subjected to heterogenous 3D refinement, non-uniform refinement, CTF refinement, and local refinement in cryoSPARC[67]. The quality of the EM density maps for the transmembrane (TM) and cytosolic domains was further improved through focused refinement, allowing for accurate model building for a major part of the protein. For the dataset of NCX1 in complex with brain PI(4,5)P_2, the maps of apo inactive NCX1 (PDB 8SGJ) and Ca^{2+}-bound active NCX1 (PDB 8SGT) are used as references for heterogeneous refinement. Due to the highly dynamic nature of the protein samples, the particles sorted into the active state produce a map with very low resolution. All resolution was reported according to the gold-standard Fourier shell correlation (FSC) using the 0.143 criterion[68]. Local resolution was estimated using cryoSPARC.

Model building, refinement, and validation

The EM maps of HsNCX1 in the SEA0400-bound and PI(4,5)P_2 diC8-bound states show high-quality density and model building is facilitated by previous apo NCX1 structure (PDB 8SGJ)[32]. Models were manually adjusted in Coot [69] and refined against maps using the phenix.real_space_refine with secondary structure restraints applied[70]. The final NCX1structural model contains residues 17-248, 370-467, 482-644, 652-698, 707-718, and 738-935. The EM map of HsNCX1 in complex with brain PI(4,5)P_2 is relatively poor. The Ca^{2+}-bound activated NCX1 structure (PDB 8SGT) is directly docked into the EM map without adjustment.

The statistics of the geometries of the models were generated using MolProbity[71]. All the figures were prepared in PyMol (Schrödinger, LLC.), Chimera[72], and ChimeraX[73].

Electrophysiological experiments

The wild-type HsNCX1 and its mutants were cloned into a pGEMHE vector and expressed in oocytes for electrophysiological recordings. RNA was synthesized using mMessage mMachine (Ambion) and injected into Xenopus laevis oocytes as described in[74]. Oocytes were isolated from at least 3 different frogs and kept at 18°C for 4-7 days. Outward HsNCX1 currents were recorded using the giant patch technique in the inside-out configuration. Each data point shown in this study represents a recording obtained from a single oocyte. The external solution (pipette solution) contained the following (mM): 100 CsOH (cesium hydroxide), 10 HEPES (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid), 20 TEAOH (tetraethyl-ammonium hydroxide), 0.2 niflumic acid, 0.2 ouabain, 8 $Ca(OH)_2$ (calcium hydroxide), pH = 7 (using MES, (2-(*N*-morpholino) ethanesulfonic acid)); bath solution (mM): 100 CsOH or 100 NaOH (sodium hydroxide), 20 TEAOH, 10 HEPES, 10 EGTA(ethylene glycol-bis(β-aminoethyl ether)-*N*,*N*,*N*',*N*'-tetra acetic acid) or HEDTA (N-(2-Hydroxyethyl) ethylenediamine-N,N',N'-triacetic acid) and different $Ca(OH)_2$ concentrations to obtain the desired final free Ca^{2+} concentrations, pH = 7 (using MES). Free Ca^{2+} concentrations were calculated according to WEBMAXc program and confirmed with a Ca^{2+} electrode.

HsNCX1 currents were evoked by the rapid replacement of 100 mM Cs^+ with 100 mM Na^+, using a computer-controlled 20-channel solution switcher. As HsNCX1 does not transport Cs^+, there is no current and only upon application of Na^+ does the exchange cycle initiate. Data were acquired at 4 ms/point and filtered at 50 Hz using an 8-pole Bessel filter. Experiments were performed at 35°C and at a holding potential of 0 mV. The effects of the Na^+-dependent inactivation were analyzed by measuring fractional currents calculated as the ratio of the steady-state current to the peak current (fractional activity). All P values were calculated using an unpaired, two-sided Welch's *t*-test.

PI(4,5)P_2 diC8 (Phosphatidylinositol 4,5-bisphosphate diC8, Echelon Bioscience) and brain PI(4,5)P_2 (L-α-phosphatidylinositol-4,5-bisphosphate, Brain, Porcine, Avanti Polar Lipids) were dissolved in water and kept as stock at −20 °C. Immediately prior to recordings, PI(4,5)P_2 was diluted in the bath solution to 10 µM final concentration and perfused cytosolically for the indicated time.

Acknowledgements

Single particle cryo-EM data were collected at the University of Texas Southwestern Medical Center Cryo-EM Facility which is funded by the CPRIT Core Facility Support Award RP170644. Cryo-EM sample grids were prepared at the Structural Biology Laboratory at UT Southwestern Medical Center which is partially supported by grant RP170644 from CPRIT. This work was

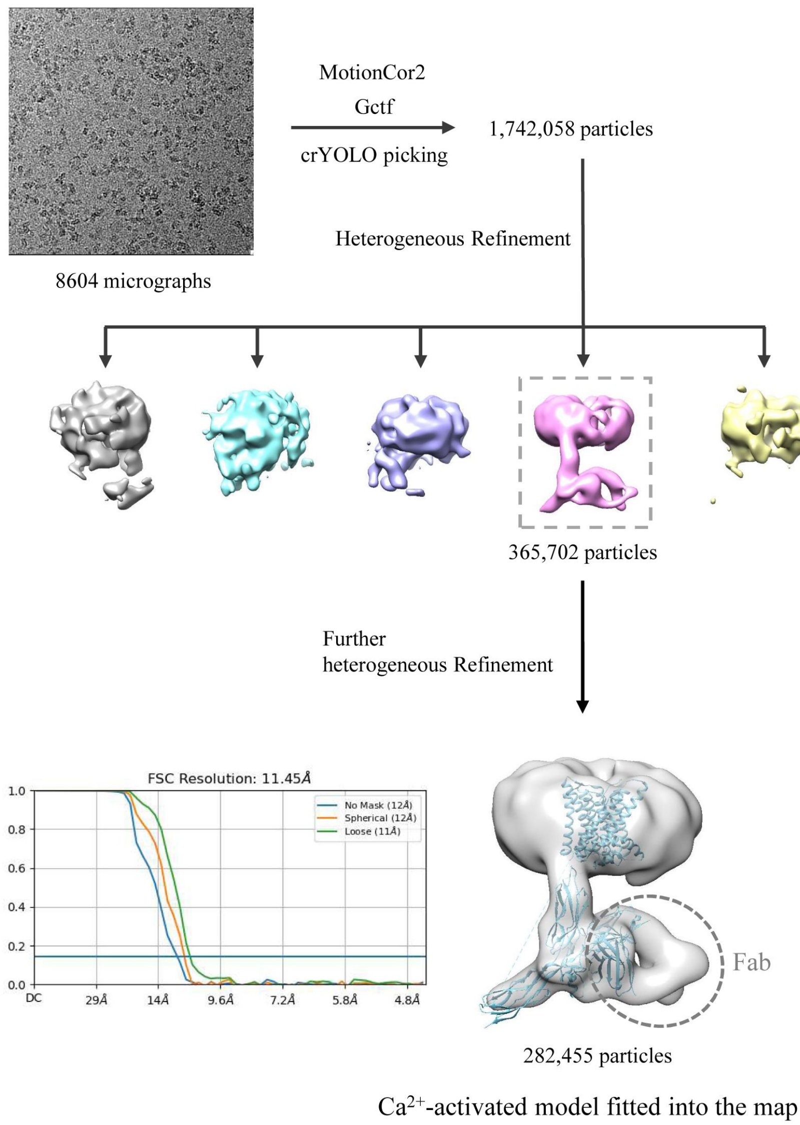

Fig. S1.

Cryo-EM data processing of HsNCX1 in the presence of long-chain porcine brain PIP_2.

The structural model of Ca2+-activated HsNCX1 from a previous study (PDB 8SGT) was directly fitted into the low-resolution EM-map (~11.5 Å). The Fab fragment from a monoclonal antibody against NCX1 was used as a fiducial marker to facilitate the single-particle alignment.

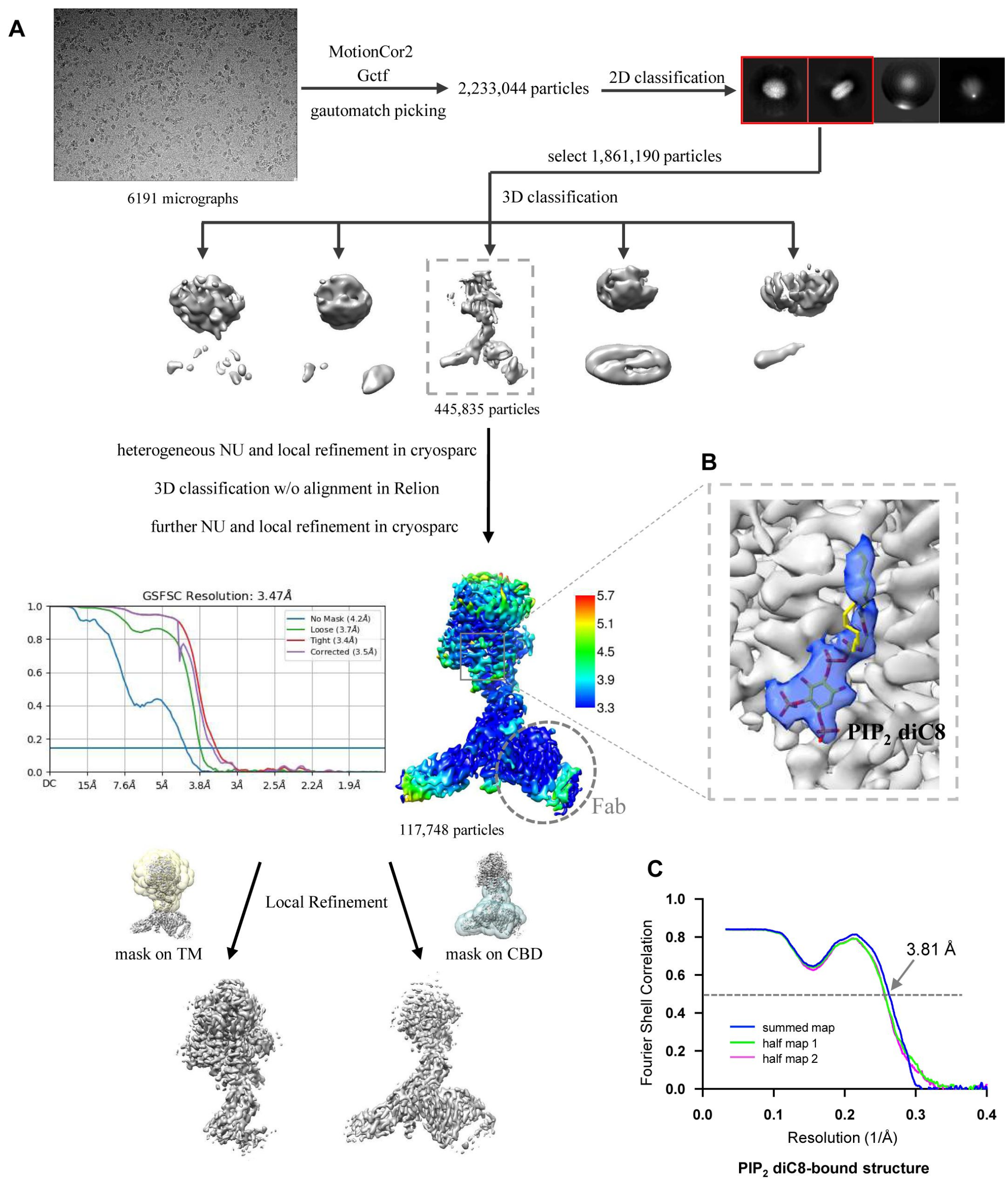

Fig. S2.

Structure determination of HsNCX1-PIP$_2$ diC8 complex.

(A) Cryo-EM data processing of HsNCX1 in complex with short-chain PIP$_2$ diC8. The Fab fragment from a monoclonal antibody against NCX1 was used as a fiducial marker to facilitate the single-particle alignment.

(B) Zoomed-in view of density map of the bound PIP$_2$ diC8 contoured at the threshold level of 0.35 using ChimeraX.

(C) The Fourier shell correlation (FSC) curves for cross-validation between the maps and the models.

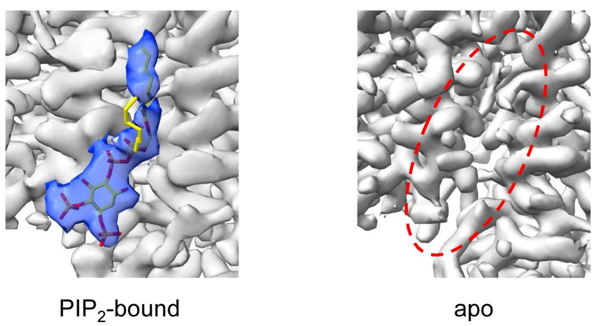

Fig. S3.

Side-by-side comparison of the densities at the PIP$_2$-binding site between the PIP$_2$-bound structure (EMD-60921) and the apo structure (EMD-40457).

The local-refined maps focused on the TM domain are used in the comparison. The density map of the bound PIP$_2$ is contoured at a threshold level of 0.35 using ChimeraX.

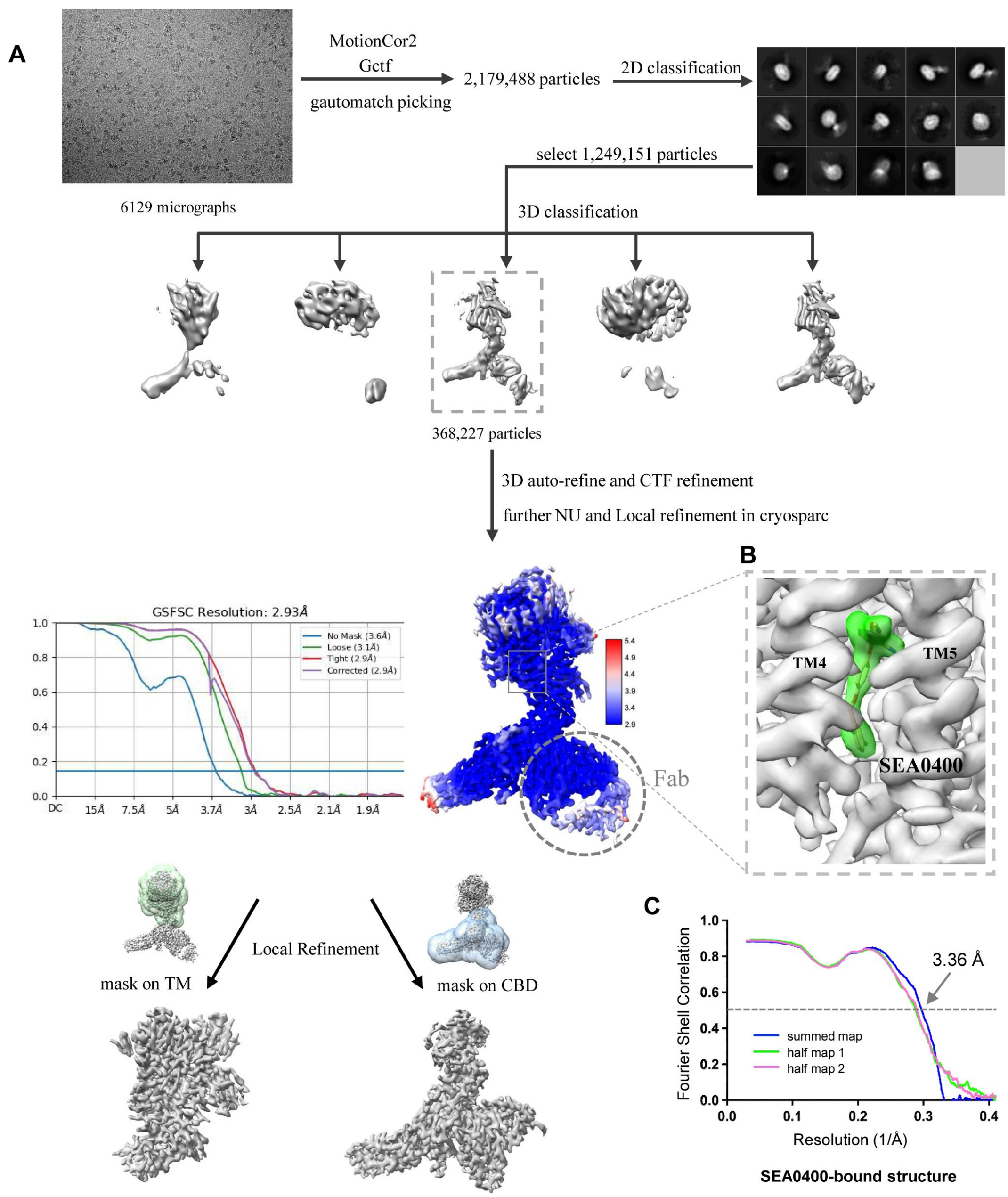

Fig. S4.

Structure determination of HsNCX1-SEA0400 complex.

(A) Cryo-EM data processing scheme of HsNCX1 in complex with SEA0400 inhibitor. The Fab fragment from a monoclonal antibody against NCX1 was used as a fiducial marker to facilitate the single-particle alignment.
(B) Zoomed-in view of density map of the bound SEA0400 inhibitor contoured at the threshold level of 0.52 using ChimeraX.
(C) The Fourier shell correlation (FSC) curves for cross-validation between the maps and the models.

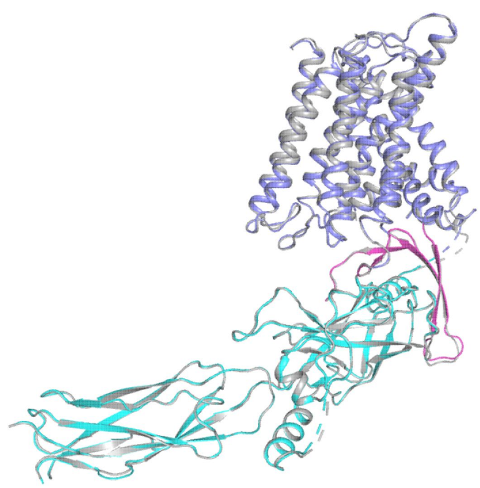

Fig. S5.

Structural comparison between the apo (grey) and SEA0400-bound (color) HsNCX1.

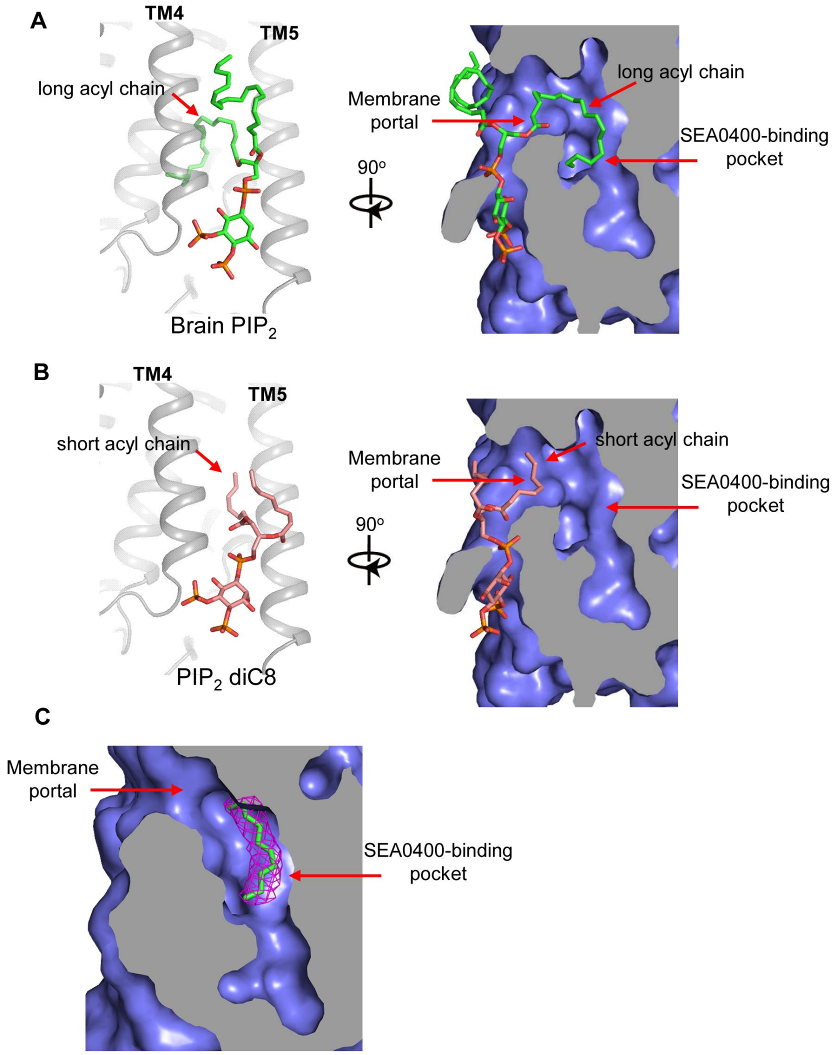

Fig. S6.

Proposed structural basis underlying the different binding affinity between long and short-chain PIP_2.

(A) A docking model for native brain PIP_2 binding in NCX1 showing the insertion of its long acyl chain into the SEA0400-binding pocket.
(B) A docking model for short-chain PIP_2 diC8 binding in NCX1 showing that the SEA0400-binding pocket is not accessible to shorter acyl chain.
(C) The density (red mesh, contoured at 6σ) of an acyl chain likely from a native lipid is observed in the SEA0400-binding pocket of the apo NCX1 structure (EMD-40457, local-refined map at the TM domain)

Sample preparation conditions	25 mM Hepes pH 7.4, 200 mM NaCl 0.9 mM SEA0400	25 mM Hepes pH 7.4, 200 mM NaCl, 0.47 mM PI(4,5)P_2 diC8
	SEA0400-bound state (EMD-40456, PDB 8SGI)	**PI(4,5)P_2 diC8-bound state** (EMD-60921, PDB 9IV8)
Data collection and processing		
Magnification	105k	105k
Voltage (kV)	300	300
Electron exposure (e^-/Å^2)	60	60
Defocus range (μm)	-0.9 - -2.2	-0.9 - -2.2
Pixel size (Å)	0.83	0.84
Symmetry imposed	C1	C1
Initial particle images (no.)	1,249,151	2,233,044
Final particle images (no.)	368,227	117,748
Map resolution (Å) FSC threshold	2.93 0.143	3.47 0.143
Refinement		
Initial model used (PDB code)	8SGJ	8SGJ
Model resolution (Å) FSC threshold	3.36 0.5	3.81 0.5
Model composition Non-hydrogen atoms Protein residues Ligands	 7667 982 3: Na 6: Ca 1: H_2O 1: SEA0400	 5947 750 5: Ca 1: PI(4,5)P_2 diC8
B factors (Å^2) Protein Ligands	 66.21 58.55	 50.47 106.20
R.m.s. deviations Bond lengths (Å) Bond angles (°)	 0.005 0.696	 0.004 0.679
Validation MolProbity score Clashscore Poor rotamers (%)	 1.36 5.24 0	 1.28 5.22 0
Ramachandran plot Favored (%) Allowed (%) Disallowed (%)	 97.62 2.38 0	 98.24 1.76 0

Table S1.

Cryo-EM data collection and model statistics.

supported in part by the Howard Hughes Medical Institute (to Y.J.) and by grants from the National Institute of Health (R35GM140892 to Y.J. and R01HL152296 to M.O.), and the Welch Foundation (Grant I-1578 to Y.J.).

Additional information

Data availability

The cryo-EM density maps of the human NCX1 have been deposited in the Electron Microscopy Data Bank under accession numbers EMD-40456 for the SEA0400-bound state and EMD-60921 for the PI(4,5)P_2 diC8-bound state, respectively. Atomic coordinates have been deposited in the Protein Data Bank under accession numbers 8SGI for SEA0400-bound structure and 9IV8 for the PI(4,5)P_2 diC8-bound structure.

Author contributions

J.X. prepared the samples and performed data acquisition, image processing, and structure determination; M.O., S.J., N.A., and W.Z. performed electrophysiology recording; Y.J. supervised the work; all authors participated in research design, data analysis, discussion, and manuscript preparation.

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Reviewer #1 (Public review):

This study uses structural and functional approaches to investigate regulation of the Na/Ca exchanger NCX1 by an activator, PIP2 and an inhibitor, SEA0400. Previous functional studies suggest both of these compounds interact with the Na-dependent inactivation process to mediate their effects.

State of the art methods are employed here, and the data are of high quality and presented very clearly. While there is merit in combining structural studies on both compounds as they relate to Na-dependent activation, in the end it is somewhat disappointing that neither is explored in further depth.

The novel aspect of this work is the study on PIP2. Unfortunately, technical limitations precluded structural data on binding of the native PIP2, and so an unnatural short-chained analog, di-C8 PIP2, was used instead. This raises the question of whether these two molecules, which have similar but very distinctly different profiles of activation, actually share the same binding pocket and mode of action. The authors conduct a "competition" experiment, arguing the effect of di-C8-PIP2 addition subsequent to PIP2 suggests competition for a single binding site. In this scenario, PIP2 would need to vacate the binding site prior to di-C8-PIP2 occupying it. However, the lack of an effect of washout alone, suggests PIP2 does not easily unbind. This raises the possibility (probability?) of a non-competitive effect of di-C8-PIP2 at a different site. An additionally informative experiment would be to determine if a saturating concentration of di-C8-PIP2 could prevent the full activation induced by subsequent PIP2 addition. However, the relative affinities of the two ligands might make such an experiment challenging in practice.

In an effort to address the binding site directly, the authors mutate key residues predicted to be important in liganding the phosphorylated head group of PIP2. However, the only mutations that have a significant effect in PIP2 activation also influence the Na-dependent inactivation process independently of PIP2. While these data are consistent with altering PIP2 binding (which cannot be easily untangled from its functional effect on Na-dependent inactivation), a primary effect on Na-inactivation, rather than PIP2 binding, cannot be fully ruled out. A more extensive mutagenic study, based on other regions of the di-C8 PIP2 binding site, would have given more depth to this work and might have been more revealing mechanistically.

The SEA0400 aspect of the work does not integrate particularly well with the rest of the manuscript. This study confirms the previously reported structure and binding site for SEA0400 but provides little further information. While interesting speculation is presented regarding the connection between SEA0400 inhibition and Na-dependent inactivation, further experiments to test this idea are not included here.

Comments on revisions:

(1) The competition assay data for di-C8-PIP2 and PIP2 is a nice addition, but in its description in the text, the authors should be a bit more circumspect about their conclusions, based on the possibility/probability that the effect observed is actually non-competitive (as detailed above).
(2) The authors should acknowledge the formal possibility that the functional effects of the mutations studies are a consequence of a direct effect on Na-dependent inactivation, independent of PIP2 binding.
(3) The authors might strengthen their arguments for combining studies on PIP2 and SEA0400.
(4) The authors could be clearer where their work on SEA0400 extends beyond the previously published observations.

https://doi.org/10.7554/eLife.105396.2.sa2

Reviewer #3 (Public review):

NCXs are key Ca2+ transporters located on the plasma membrane, essential for maintaining cellular Ca2+ homeostasis and signaling. The activities of NCX are tightly regulated in response to cellular conditions, ensuring precise control of intracellular Ca2+ levels, with profound physiological implications. Building upon their recent breakthrough in determining the structure of human NCX1, the authors obtained cryo-EM structures of NCX1 in complex with its modulators, including the cellular activator PIP2 and the small molecule inhibitor SEA0400. Structural analyses revealed mechanistically informative conformational changes induced by PIP2 and elucidated the molecular basis of inhibition by SEA0400. These findings underscore the critical role of the interface between the transmembrane and cytosolic domains in NCX regulation and small molecule modulation. Overall, the results provide key insights into NCX regulation, with important implications for cellular Ca2+ homeostasis.

Comments on revisions:

The authors have adequately addressed my previous comments.

https://doi.org/10.7554/eLife.105396.2.sa1

Author response:

The following is the authors’ response to the original reviews

> ***Reviewer #1 (Public Review):***
>
> *(1) This study uses structural and functional approaches to investigate the regulation of the Na/Ca exchanger NCX1 by an activator, PIP2, and an inhibitor, SEA0400. State-of-the-art methods are employed, and the data are of high quality and presented very clearly. The manuscript combines two rather different studies (one on PIP2; and one on SEA0400) neither of which is explored in the depth one might have hoped to form robust conclusions and significantly extend knowledge in the field.*

We combined the study of PIP2 and SEA0400 in this manuscript because both ligands inhibit or activate NCX1 by affecting the Na^+-dependent inactivation of the exchanger - SEA0400 promotes inactivation by stabilizing the cytosolic inactivation assembly whereas PIP2 mitigates inactivation by destabilizing the assembly. The current study aims to provide structural insights into these ligand binding. We didn't perform extensive electrophysiological analysis as the functional effects of both ligands have been extensively characterized over the last thirty years.

> *(2) The novel aspect of this work is the study of PIP2. Unfortunately, technical limitations precluded structural data on binding of the native PIP2, so an unnatural short-chained analog, diC8 PIP2, was used instead. This raises the question of whether these two molecules, which have similar but very distinctly different profiles of activation, actually share the same binding pocket and mode of action. In an effort to address this, the authors mutate key residues predicted to be important in forming the binding site for the phosphorylated head group of PIP2. However, none of these mutations prevent PIP2 activation. The only ones that have a significant effect also influence the Na-dependent inactivation process independently of PIP2, thus casting doubt on their role in PIP2 binding, and thus identification of the PIP2 binding site. A more extensive mutagenic study, based on the diC8 PIP2 binding site, would have given more depth to this work and might have been more revealing mechanistically.*

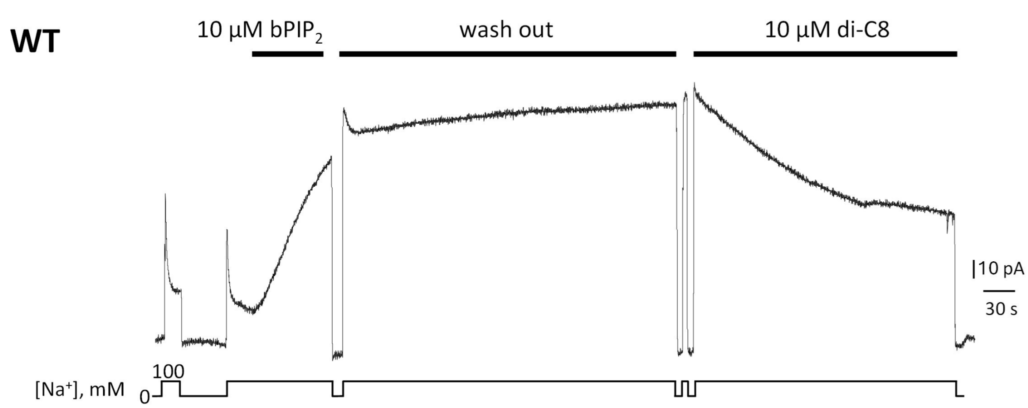
The reviewer raises the important question of whether the short-chain PIP2 diC8 and long-chain native PIP2 share the same binding site. We have performed a pilot experiment to address this question. The data indicate that PIP2 diC8 competes with native brain PIP2 for its binding site (Author response image 1). We believe that the mild effects of diC8 on the biophysical properties of NCX1 are due to its decreased affinity as compared to the long-chain PIP2. We have included this competition assay in the revised manuscript.

The acyl-chain length-dependent PIP2 activation is consistent with some previous studies. Before PIP2 was demonstrated to regulate NCX1, some earlier studies showed that negatively charged long-chain lipids such as phosphatidylserine (PS) or phosphatidic acid (PA) could have the same potentiation effects on NCX1 as PIP2 (PMID: 1474504; PMID: 3276350). A later study showed that long-chain acyl-CoAs could also have the same potentiation effects on NCX1 as PIP2 (PMID: 16977318). All these studies demonstrated that activation of NCX by the anionic lipids depends on their chain length with the short chain being ineffective or less effective. These findings have two implications. First, it is the negative surface charge rather than the specific IP3 head group of the lipid that is important for stimulating NCX1 activity. This would imply non-specific electrostatic interactions between the negatively charged lipids and those positively charged residues at the binding site. Second, a longer acyl chain is required for the high-affinity binding of PIP2 or negatively charged lipids. As further discussed in the revised manuscript (Discussion section), we suspect the tail of the long acyl chain from the native anionic lipids can enter the same binding pocket for SEA0400 thereby rendering higher affinity lipid binding than shorter chain lipids.

As the interactions between PIP2 and NCX1 are both electrostatic involving multiple charged residues as well as hydrophobic involving the long lipid acyl chain, single amino acid substitutions likely only decrease the affinity of PIP2 rather than completely disrupt its binding. Our data demonstrated that mutants R220A, K225A, and R220A/K225A do show a significantly decreased potentiation effect of PIP2 (Figure 3 in the manuscript). We also conducted an experiment with a mutant exchanger in which all four amino were mutated. This K164A/R167A/R220A/K225A mutant is insensitive to PIP2 and shows no Na^+-dependent inactivation (Figure 3A). The unresponsiveness to PIP2 and lack of Na^+-dependent inactivation in this mutant is consistent with previous studies demonstrating that PIP2 activates NCX by tuning the amount of Na^+-dependent inactivation and any mutation that decreases NCX sensitivity to PIP2 will affect the extent of Na^+-dependent inactivation (PMID: 10751315). Such studies show that the two processes cannot be dissected from each other, making more extensive mutagenesis investigation unlikely to provide new mechanistic insights. A brief discussion related to this quadruple mutant has been added in the revised manuscript.

Author response image 1.

Giant patch recording of the human WT exchanger. Currents were first activated by intracellular application of 10 µM brain PIP2. Afterwards, a solution containing 100 mM Na^+ and 12 µM Ca^{2+} was perfused for about 5 min (washout). The PIP2 effects was not reversible during this time. The same patch was then perfused internally with the same solution in presence of 10 µM di-C8. Application of the shorted-chained di-C8, partially decreased the current suggesting that that PIP2 and diC8 compete for the binding site.



> *(3) The SEA0400 aspect of the work does not integrate particularly well with the rest of the manuscript. This study confirms the previously reported structure and binding site for SEA0400 but provides no further information. While interesting speculation is presented regarding the connection between SEA0400 inhibition and Na-dependent inactivation, further experiments to test this idea are not included here.*

Our SEA0400-bound NCX structure was determined and deposited in 2023, along with our previous study on the apo NCX published in 2023 (PMID: 37794011). We decided to combine the SEA0400-bound structure with the later study of PIP2 binding because both represent ligand modulation of NCX by affecting the Na^+-dependent inactivation of the exchanger. The SEA0400 inhibition of NCX1 has been extensively investigated previously, which demonstrated a strong connection between SEA0400 and the Na^+-dependent inactivation. As discussed in the manuscript, SEA0400 is ineffective in an exchanger lacking Na^+-dependent inactivation. Conversely, enhancing the extent of Na^+-dependent inactivation increases the affinity for SEA0400. Our structural analysis provides explanations for these pharmacological features of SEA0400 inhibition.

> ***Reviewer #2 (Public review):***
>
> *(1) The study by Xue et al. reports the structural basis for the regulation of the human cardiac sodium-calcium exchanger, NCX1, by the endogenous activator PIP2 and the small molecule inhibitor SEA400. This well-written study contextualizes the new data within the existing literature on NCX1 and the broader NCX family. This work builds upon the authors' previous study (Xue et al., 2023), which presented the cryo-EM structures of human cardiac NCX1 in both inactivated and activated states. The 2023 study highlighted key structural differences between the active and inactive states and proposed a mechanism where the activity of NCX1 is regulated by the interactions between the ion-transporting transmembrane domain and the cytosolic regulatory domain. Specifically, in the inward-facing state and at low cytosolic calcium levels, the transmembrane (TM) and cytosolic domains form a stable interaction that results in the inactivation of the exchanger. In contrast, calcium binding to the cytosolic domain at high cytosolic calcium levels disrupts the interaction with the TM domain, leading to active ion exchange.*
>
> *In the current study, the authors present two mechanisms explaining how both PIP2 stimulates NCX1 activity by destabilizing the protein's inactive state (i.e., by disrupting the interaction between the TM domain and the cytosolic domain) and how SEA400 stabilizes this interaction, thereby acting as a specific inhibitor of the system.*
>
> *The first part of the results section addresses the effect of PIP2 and PIP2 diC8 on NCX1 activity. This is pertinent as the authors use the diC8 version of this lipid (which has a shorter acyl chain) in their subsequent cryo-EM structure due to the instability of native PIP2. I am not an electrophysiology expert; however, my main comment would be to ask whether there is sufficient data here to characterise fully the differences between PIP2 and PIP2 diC8 on NCX1 function. It appears from the text that this study is the first to report these differences, so perhaps this data needs to be more robust. The spread of the data points in Figure 1B is possibly a little unconvincing given that only six measurements were taken. Why is there one outlier in Figure 1A? Were these results taken using the same batch of oocytes? Are these technical or biological replicates? Is the convention to use statistical significance for these types of experiments?*

Oocytes were isolated from at least 3 different frogs and each data point shown in Fig. 1 A or 1B of the manuscript represents a recording obtained from a single oocyte. For clarity, we have added this information to the Methods section. We understand that 6 observations (Fig. 1B) are a small sample size but electrophysiological recordings of NCX currents are extremely challenging and technically difficult due to the low transport activity of the exchanger. Because of these circumstances, this type of study relies on a small sample of observations. Nevertheless, our data clearly show that native PIP2 and the short-chain PIP2 diC8 can activate NCX activity although with different affinity. The spread of the steady state current data points is due to the variability in the extent of Na^+-dependent inactivation within each patch, likely due to slightly different levels of endogenous PIP2 or other regulatory mechanisms that control this allosteric process. As PIP2 acts on the Na^+-dependent inactivation this will lead to varying levels of potentiation. Because of that, we did occasionally observe some outliers in our recordings. Rather than cherry-picking in data analysis, we presented all the data points from patches with measurable NCX1 currents. Despite this variability, a T-test indicates that the effects of PIP2 are more pronounced on the steady-state current than peak current. The differences between native PIP2 and PIP2 diC8 on NCX1 function are consistent with previous investigations showing that both PIP2 and anionic lipids enhance NCX current by antagonizing the Na^+-dependent inactivation and long-chain lipids are more effective in potentiating NCX1 activity (PMID: 1474504; PMID: 3276350; PMID: 16977318). A discussion related to the chain length-dependent lipid activation of NCX1 is added in the Discussion of the revised manuscript.

> *(2) I am also somewhat skeptical about the modelling of the PIP2 diC8 molecule. The authors state, "The density of the IP3 head group from the bound PIP2 diC8 is well-defined in the EM map. The acyl chains, however, are flexible and could not be resolved in the structure (Fig. S2)."*
>
> *However, the density appears rather ambiguous to me, and the ligand does not fit well within the density. Specifically, there is a large extension in the volume near the phosphate at the 5' position, with no corresponding volume near the 4' phosphate. Additionally, there is no bifurcation of the volume near the lipid tails. I attempted to model cholesterol hemisuccinate (PDB: Y01) into this density, and it fits reasonably well - at least as well as PIP2 diC8. I am also concerned that if this site is specific for PIP2, then why are there no specific interactions with the lipid phosphates? How can the authors explain the difference between PIP2 and PIP2 diC8 if the acyl chains don't make any direct interactions with the TM domain? In short, the structures do not explain the functional differences presented in Figure 1.*
>
> *The side chain densities for Arg167 and Arg220 are also quite weak. While there is some density for the side chain of Lys164, it is also very weak. I would expect that if this site were truly specific for PIP2, it should exhibit greater structural rigidity - otherwise, how is this specific?*
>
> *Given this observation, have the authors considered using other PIP2 variants to determine if the specificity lies with $PI4,5P_2$ as opposed to $PI3,5P_2$ or $PI3,4P_2$? A lack of specificity may explain the observed poor density.*

The map we provided to the editor in the initial submission is the overall map for PIP2-bound NCX1. Due to the relative flexibility between the cytosolic CBD and TM regions, we also performed local refinement on each region in data processing to improve the map quality as illustrated in Fig. S2. The local-refined map focused on the TM domain provides a much better density for PIP2 diC8 and its surrounding residues than the overall map. The map quality allowed us to unambiguously identify the lipid as PIP2 with the IP3 head group having phosphate groups at the 4,5 positions. Furthermore, no lipid density is observed at the equivalent location in the local-refined map from the apo NCX1 TM region as shown in Fig. S3 in the revision. In the revised manuscript, the density for the bound PIP2 is shown in Fig. 2A. Those local-refined maps for PIP2-bound NCX1 were also deposited as additional maps along with the overall map in the Electron Microscopy Data Bank under accession numbers EMD-60921. The local-refined maps for the apo-NCX1 were deposited in the Electron Microscopy Data Bank under accession numbers EMD-40457 in our previous study (https://www.ebi.ac.uk/emdb/EMD-40457?tab=interpretation).

As discussed in our response to reviewer #1, the acyl-chain length-dependent PIP2 activation is consistent with some previous studies. Before PIP2 was identified as a physiological regulator of NCX1, some earlier studies showed that negatively charged long-chain lipids such as phosphatidylserine (PS) or phosphatidic acid (PA) could have the same potentiation effects on NCX as PIP2 (PMID: 1474504; PMID: 3276350). A later study also showed that acyl-CoA could also have the same potentiation effects on NCX as PIP2 (PMID: 16977318). All these studies demonstrated that activation of NCX1 by the anionic lipids depends on their chain length with the short chain being ineffective. These findings have two implications. First, it is the negative surface charge rather than the specific IP3 head group of the lipid that is important for stimulating NCX activity. This would imply non-specific electrostatic interactions between the negatively charged lipids and those positively charged residues at the binding site. Second, a longer acyl chain is required for the high-affinity binding of PIP2 or negatively charged lipids. As further discussed in the revised manuscript (Discussion section), we suspect the tail of the long acyl chain can enter the same binding pocket for SEA0400 thereby rendering higher affinity lipid binding than shorter chain lipids. In light of the equivalent potentiating effect of various anionic lipids on NCX1, PI(4,5)P2 activation of NCX1 is likely non-specific and PI(3,5)P2 or PI(3,4)P2 may also activate the exchanger. However, as a key player in membrane signaling, PI(4,5)P2 has been demonstrated to be a physiological regulator of NCX1 in many studies.

> *(3) I also noticed many lipid-like densities in the maps for this complex. Is it possible that the authors overlooked something? For instance, there is a cholesterol-like density near Val51, as well as something intriguing near Trp763, where I could model PIP2 diC8 (though this leads to a clash with Trp763). I wonder if the authors are working with mixed populations in their dataset. The accompanying description of the structural changes is well-written (assuming it is accurate).*

Densities from endogenous lipids and cholesterols are commonly observed in membrane protein structures. Other than the bound PIP2, those lipid and cholesterol densities are present in both the apo and PIP2-bound structures, including the density around Trp763 and Val53. Whether those bound lipids/cholesterols play any functional roles or just stabilize the protein is beyond the scope of this study. We have added a supporting figure (Fig. S3) showing a side-by-side comparison of the density at the PIP2 binding site between the PIP2-bound and apo structures.

> *I would recommend that the authors update the figures associated with this section, as they are currently somewhat difficult to interpret without prior knowledge of NCX architecture. My suggestions include:*
>
> *- Including the density for the PIP2 diC8 in Figure 2A.*

As suggested, we have included the density of PIP2 diC8 in Figure 2A.

> *- Adding membrane boundaries (cytosolic vs. extracellular) in Figure 2B.*
>
> *- Labeling the cytosolic domains in Figure 2B.*
>
> *- Adding hydrogen bond distances in Figure 2A.*

We have added and labeled the boundaries for the TM and cytosolic domains in Figure 2B as suggested. Although we can identify those positively charged residues in the vicinity of the PIP2 head group and observe local structural changes, the poorly defined side-chain densities of these residues won't allow us to properly determine the hydrogen bond distances.

> *- Detailing the domain movements in Figure 2B (what is the significance of the grey vs. blue structures?).*

There is a rigid-body downward swing movement at CBDs between the apo (grey) and PIP2-bound (cyan) structures. The movement at the TM region is subtle. We have added the description in the legend for Figure 2B and also marked the movement at the tip of CBD1 in the figure.

> *The section on the mechanism of SEA400-induced inactivation is strong. The maps are of better quality than those for the PIP2 diC8 complex, and the ligand fits well. However, I*

> *noticed a density peak below F02 on SEA400 that lies within the hydrogen bonding distance of Asp825. Is this a water molecule? If so, is this significant?*

The structure of SEA0400-bound NCX1 was determined at a higher resolution likely because the drug stabilize the exchanger in the inactivated state. The mentioned density could be an ordered water molecule. We don't know if it is functionally significant.

> *Furthermore, there are many unmodeled regions that are likely cholesterol hemisuccinate or detergent molecules, which may warrant further investigation.*

We constantly observed partial densities from bound lipids, cholesterols, or detergents in our structures. Most of them are difficult to be unambiguously identified and modeled. Whether they play any functional roles is beyond the scope of this study.

> *The authors introduce SEA400 as a selective inhibitor of NCX1; however, there is little to no comparison between the binding sites of the different NCX proteins. This section could be expanded. Perhaps Fig. 4C could include sequence conservation data.*

SEA0400 is more specific for NCX1 than NCX2 and NCX3 as demonstrated in an early study (PMID: 14660663). The lack of structure information for NCX2 or NCX3 makes it difficult to make a direct comparison to reveal the structural basis of SEA0400 specificity.

> *Additionally, is the fenestration in the membrane physiological, or is it merely a hole forced open by the binding of SEA400? I was unclear as to whether the authors were suggesting a physiological role for this feature, similar to those observed in sodium channels.*

The fenestration likely serves as the portal for SEA0400 binding as discussed in the manuscript. As further discussed in the revised manuscript, we suspect this fenestration also allows the tail of a long-chain lipid to enter the same binding pocket for SEA0400 and results in higher affinity binding of a long-chain lipid than a short-chain lipid.

> ***Reviewer #3 (Public review):***
>
> *NCXs are key Ca^{2+} transporters located on the plasma membrane, essential for maintaining cellular Ca^{2+} homeostasis and signaling. The activities of NCX are tightly regulated in response to cellular conditions, ensuring precise control of intracellular Ca^{2+} levels, with profound physiological implications. Building upon their recent breakthrough in determining the structure of human NCX1, the authors obtained cryo-EM structures of NCX1 in complex with its modulators, including the cellular activator PIP2 and the small molecule inhibitor SEA0400. Structural analyses revealed mechanistically informative conformational changes induced by PIP2 and elucidated the molecular basis of inhibition by SEA0400. These findings underscore the critical role of the interface between the transmembrane and cytosolic domains in NCX regulation and small molecule modulation. Overall, the results provide key insights into NCX regulation, with important implications for cellular Ca^{2+} homeostasis.*

We appreciate this reviewer's positive comments.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *The manuscript would be strengthened enormously by a much deeper focus on the novel and very interesting PIP2 work, as noted above, and perhaps the removal of the SEA0400*

> *data.*
>
> *If that is beyond the scope of the authors' options, then a more robust discussion of limitations of the current work, perhaps speculation regarding other future experiments, a clearer presentation of how these data on SEA0400 are different from/extend from the previously published work, and a better effort to link the two disparate aspects of the work into a more cohesive manuscript should be attempted.*

As discussed in our response to this reviewer's public review, we combined the study of PIP2 and SEA0400 in this manuscript because both ligands activate or inhibit NCX1 by affecting the Na^+-dependent inactivation of the exchanger. The functional effects of both ligands on NCX1 have been extensively characterized over the last thirty years. Thus the current study is focused on providing structural explanations for some unique pharmacological features of these ligands. In the revised manuscript, we have added an extra paragraph of discussion that provides a plausible explanation for chain length-dependent PIP2 activation.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *A few comments to consider:*
>
> *(1) The short-chain PIP2 appears to have lower potency, but the mechanism remains unclear. Based on structural analyses, are there potential binding sites for the acyl chains of PIP2 that could contribute to this difference?*

As discussed in our response to other reviewers, long-chain anionic lipids can have the same potentiation effect on NCX1 activity as PIP2, but the short-chain ones are ineffective just like short-chain PIP2 diC8. We suspect the tail of a long acyl chain from the native PIP2 can enter the same binding pocket for SEA0400 thereby rendering higher affinity binding for a long-chain lipid than a short-chain lipid. A discussion related to this point has been added to the revised manuscript.

> *(2) It is unclear why mutating residues that interact with the IP3 head group retain PIP2 activation. Would it be possible to assess PIP2 and C8 PIP2 binding to these NCX1 variants? Identifying a mutant that abolishes C8 PIP2 binding would be valuable in interpreting those results.*

As the interactions between PIP2 and NCX1 are both electrostatic involving multiple charged residues and hydrophobic involving the long lipid acyl chain, single amino acid substitutions likely only decrease the affinity of PIP2 rather than completely disrupt its binding. Individual mutants R220A and K225A show a 5-fold decrease in their response to PIP2 application indicating that their replacement alters the affinity of NCX for PIP2. We have added a new experiment showing that an exchanger with all four residues mutated is insensitive to PIP2 in the revision.

> *(3) What are the functional effects of mutating Y226 and R247, residues that seem to play an important role in PIP2-mediated activation?*

In a previous study, mutation at Y226 (Y226T), which is found within the XIP region of NCX, has been shown to have enhanced Na^+-dependent inactivation (PMID: 9041455). To our knowledge, the R247 mutation has not been investigated. Also positioned in the XIP region, we suspect its mutation could directly affect Na^+-dependent inactivation. This would make it difficult to determine if the function effect of the mutation is caused by changing the stability of the XIP region or by changing the binding of PIP2.

> *(4) Is there any overlap between the PIP2 and SEA0400 binding regions? Both appear to involve TM4, TM5, and TMD-beta hub interfaces. It might be interesting to discuss any shared mechanisms and why this region might serve as a hotspot for modulation.*

As mentioned in our previous response, we suspect the tail of a long acyl chain from the native PIP2 can enter the same binding pocket for SEA0400 thereby rendering higher affinity binding for a long-chain lipid than a short-chain lipid. A more detailed discussion related to this point has been included in the revision.

> *(5) It would be helpful to show the density at the PIP2-binding site in the apo and PIP2-bound structures side by side*

This figure has been added in the revision as Fig. S3.

https://doi.org/10.7554/eLife.105396.2.sa0